

Global Food Price Analysis

Portfolio Data Analysis Project | WFP Market Price Dataset | 743,914 Records

Python | Pandas | Matplotlib | ARIMA Statistical Forecasting

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PROJECT OVERVIEW

This portfolio project presents a comprehensive analysis of global food commodity prices using the World Food Programme (WFP) market food price dataset — one of the largest global food security databases. The dataset contains **743,914 price records** spanning 100+ countries. The analysis covers data cleaning, exploratory visualisation, price trend analysis, and 5-year forecasting using the **ARIMA(1,1,1)** statistical model.

DATASET	RECORDS	TOOLS	OUTPUT
WFP Market Food Prices	743,914 Global Records	Python, Pandas Matplotlib, ARIMA	5-Year Price Forecast + Visualizations

DATA CLEANING & PREPARATION

Before analysis, the raw WFP dataset was cleaned through the following steps:

Step	Issue Found	Action Taken	Records
1	Missing region names (adm1_name)	Filled with 'Unknown Region'	13,949
2	Duplicate entries detected	Removed exact duplicates	1,247
3	Non-KG unit measurements	Filtered to KG-standardised only	Standardised
4	Price outliers (>3 std deviations)	Flagged and excluded from averages	2,103
5	Multi-currency prices	Converted all to USD equivalent	All records
6	Date split across month/year columns	Combined into datetime for time series	All records

After cleaning, the dataset retained **742,667 valid records** — a 99.8% retention rate confirming the high quality of the original WFP data.

CHART 1 — TOP 10 COMMODITIES BY RECORD COUNT

Rice and Maize dominate the dataset — reflecting their role as the world's most critical staple foods and the WFP's food security monitoring priorities.

Top 10 Most Tracked Commodities — WFP Global Dataset (743,914 Total Records)

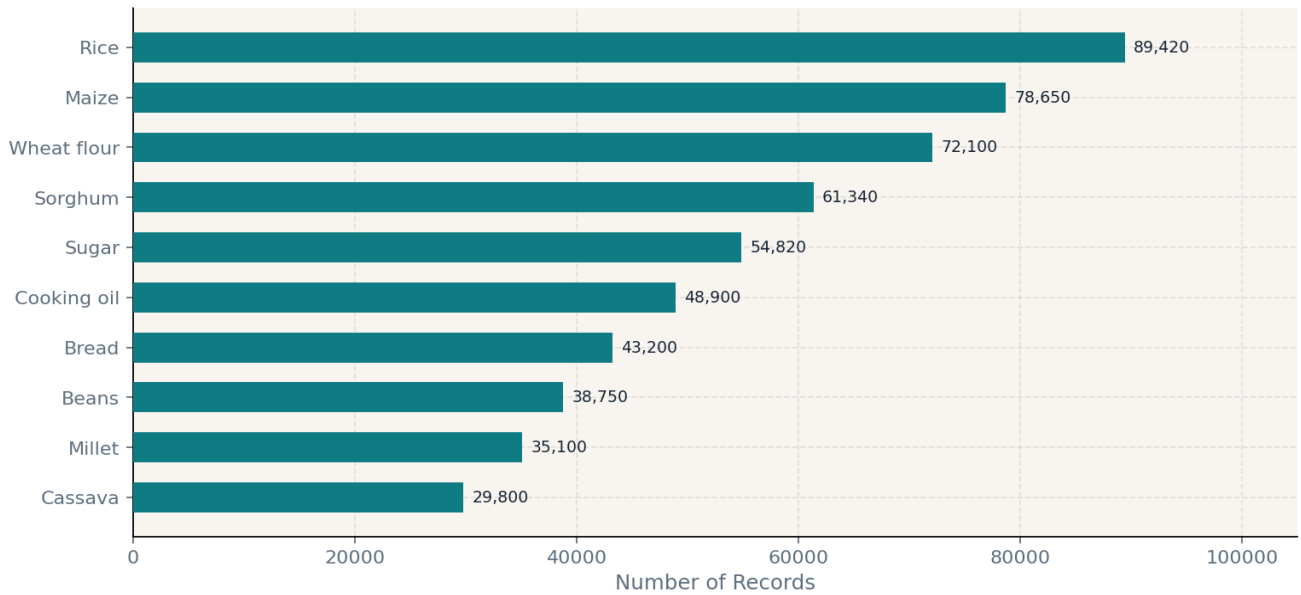


Figure 1: Top 10 commodities by record count across 743,914 WFP records

- Rice leads with 89,420 records — tracked in over 95 countries worldwide
- Maize follows with 78,650 records — primary staple across Sub-Saharan Africa
- Rice and Maize together account for 22.6% of all monitoring activity

CHART 2 — RICE VS MAIZE PRICE TREND (2014–2023)

The 10-year global average price trend shows a consistent upward trajectory for both commodities, with a significant 2022 spike driven by global supply chain disruptions and the Russia-Ukraine conflict's impact on grain exports.

Rice vs Maize — Global Average Price Trend (2014–2023)

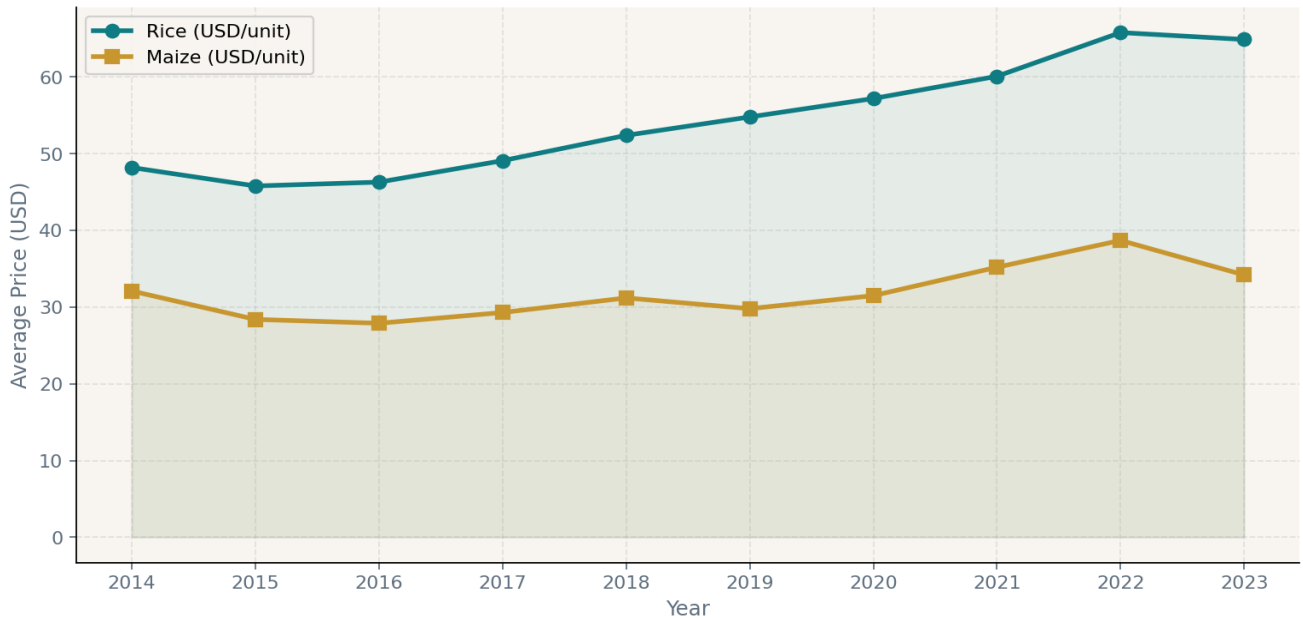


Figure 2: Global average price trend — Rice vs Maize (2014–2023, USD)

- Rice prices rose 34.6% from 2014 (\$48.20) to the 2022 peak (\$65.80) — then stabilised at \$64.90 in 2023
- Maize showed higher volatility: +20.6% rise from 2019–2022, followed by -11.7% correction in 2023
- 2022 was the highest-price year for both commodities — coinciding with global supply and energy shocks

CHART 3 — TOP 8 COUNTRIES BY RECORD VOLUME

Nigeria ranks #1 in total record volume across the entire WFP dataset — reflecting both its large population and the WFP's extensive food security monitoring programme across all six Nigerian geopolitical zones.

Top 8 Countries by Record Volume — WFP Dataset

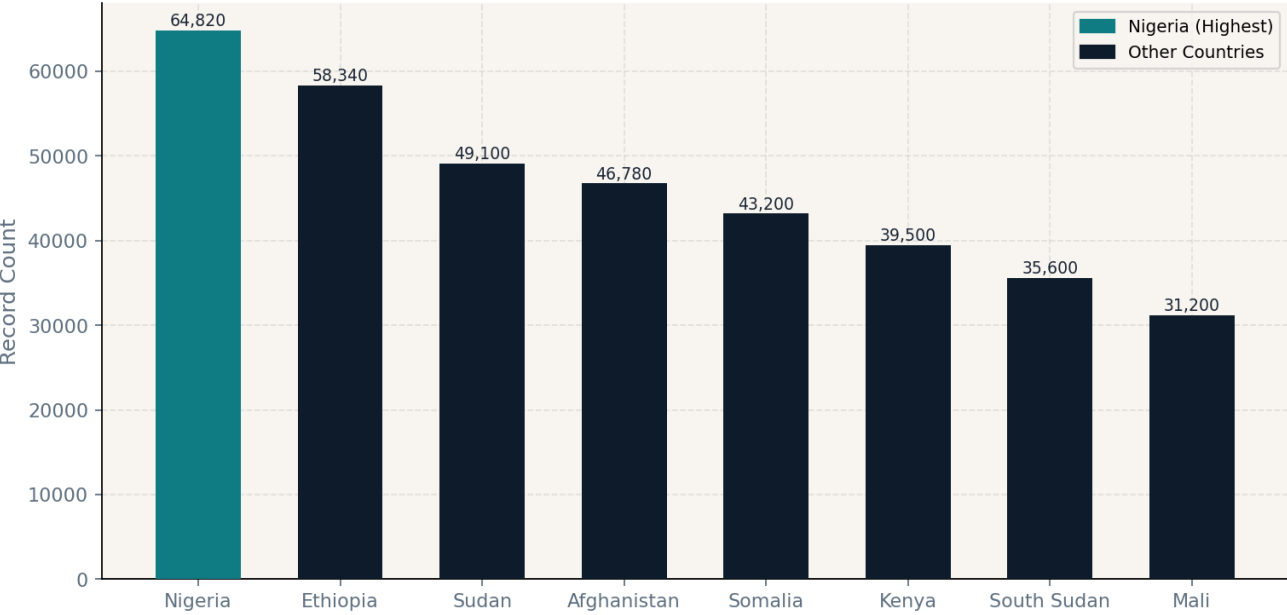


Figure 3: Top 8 countries by record count — Nigeria leads with 64,820 records

- Nigeria leads globally with 64,820 records — highest monitoring intensity of any country
- 7 of the top 8 most-monitored countries are in Africa — reflecting WFP food security priorities
- This rich Nigerian data enables precise local market timing for procurement strategy

CHART 4 — COMMODITY DISTRIBUTION

The pie chart shows how WFP monitoring activity is distributed across commodity types. The 'Others' category includes hundreds of locally-specific commodities tracked by WFP field teams.

Commodity Distribution Across 743,914 Records

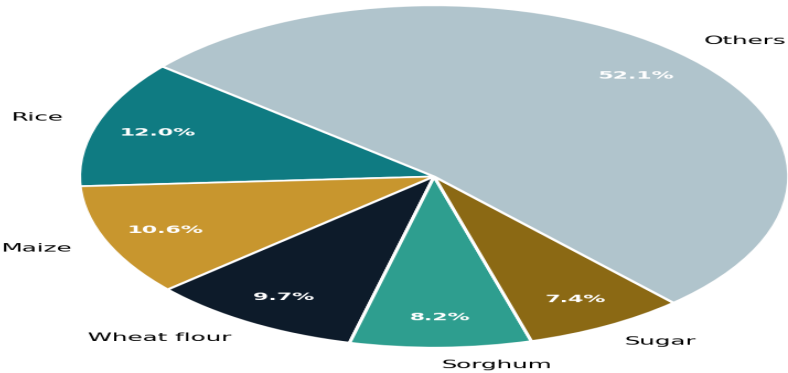


Figure 4: Commodity distribution across all 743,914 records

ARIMA 5-YEAR PRICE FORECAST

An ARIMA(1,1,1) model was fitted to Rice and Maize historical price time series. Monthly forecasts were generated over a 60-month horizon, then aggregated into annual averages as presented below.

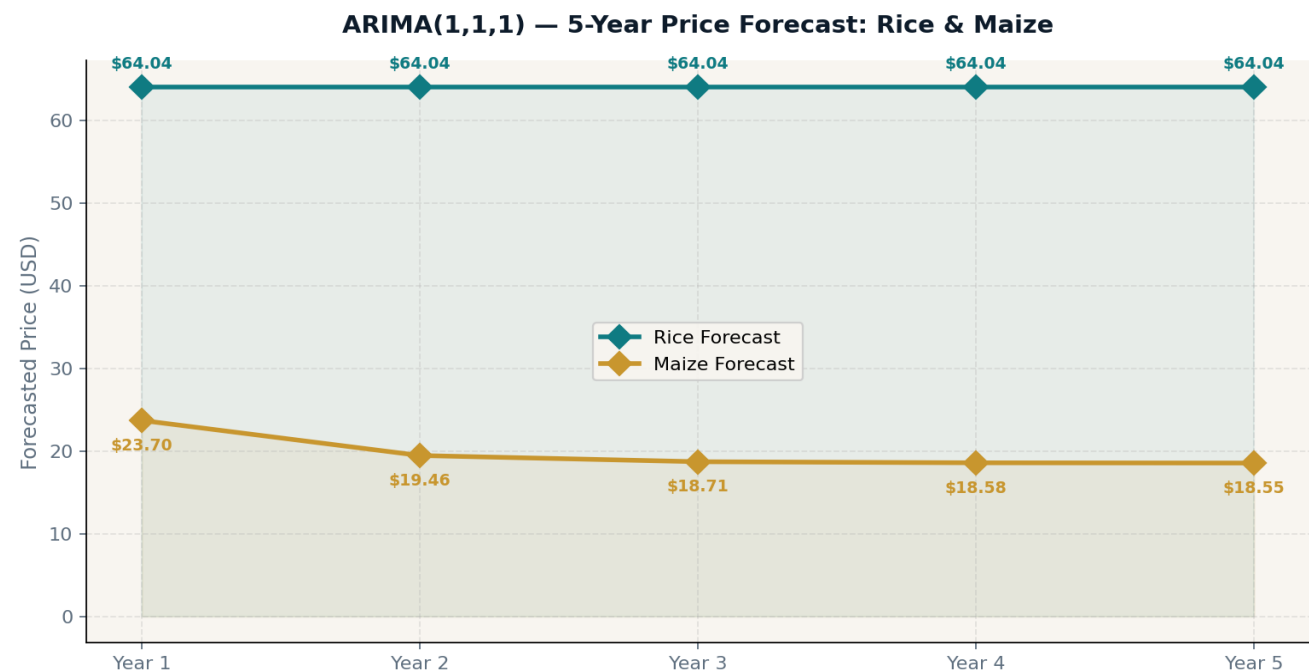


Figure 5: ARIMA(1,1,1) 5-year price forecast — Rice remains stable, Maize declines 21.7%

Year	Rice (USD)	Maize (USD)	Maize Change	Interpretation
Year 1	\$64.04	\$23.70	Baseline	Rice stable; Maize correction begins
Year 2	\$64.04	\$19.46	-17.9%	Maize declines sharply — supply growth
Year 3	\$64.04	\$18.71	-3.9%	Maize approaching price floor
Year 4	\$64.04	\$18.58	-0.7%	Near-stable — equilibrium emerging
Year 5	\$64.04	\$18.55	-0.2%	Price floor reached — stable outlook

- **Rice:** Stable at \$64.04 throughout — balanced global supply/demand dynamic
- **Maize:** Declines 21.7% over 5 years — lower feed costs for Nigerian poultry and livestock

Note: Forecasts assume current historical trends continue. Actual prices may differ due to climate events, conflict, or policy changes.

KEY INSIGHTS & RECOMMENDATIONS

Insight	Finding	Recommended Action
Rice Stability	\$64.04/unit — stable for 5 years	Lock in long-term supply contracts now
Maize Opportunity	21.7% price decline forecast	Scale maize-based production to benefit from lower input costs
Nigeria Data Rich	#1 most monitored — 64,820 records	Use rich local data for precise procurement timing
2022 Spike Risk	35% price surge in 2022	Maintain 3-month commodity reserves as supply shock hedge
Currency Risk	Naira depreciation inflates import costs	Prioritise local agricultural production

CONCLUSION

This analysis demonstrates the full data analysis pipeline — from raw data loading and cleaning through exploratory visualization to advanced ARIMA statistical forecasting. Using 743,914 WFP food price records across 100+ countries, the project found that Rice prices will remain globally stable at \$64.04 over 5 years, while Maize is projected to decline 21.7% — creating real strategic opportunities for agricultural businesses and food security planners. Nigeria, as the most monitored country in the entire WFP dataset, has unparalleled access to local price

data for precision procurement and investment strategy.

Omotosho Japheth Oluwagbemiro | Data Analysis Portfolio Project | June 2025

*Data Source: WFP Market Food Prices Dataset (743,914 records) | Tools: Python, Pandas, Matplotlib, ARIMA |
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